Group 3

BDM 3035 - Big Data Capstone

CAMEL-AI Research Paper Milestone 3

Presented by:

Chaw Su Su Thinn

Date:

2025-07-08

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Why This Research?

THE PROBLEM - LLMs LIMITATIONS

- Users spend 40-60% of time crafting prompts
- Domain expertise creates barriers
- Need constant human guidance

THE FUTURE - AGENT SOCIETY

- Agents plan tasks & exchange ideas
- Collaborate problem-solving like human teams
- Minimal input to complete complex tasks

RESEARCH QUESTION

"Can AI agents communicate and cooperate to solve tasks with minimal intervention?"

CAMEL-AI Overview

What is CAMEL-AI ?

- ❏ CAMEL = Communicative Agents for "Mind" Exploration of LLMs.
- ❏ Explores how AI agents can collaborate, reason, and solve tasks autonomously, with minimal human input.
- ❏ Focuses on multi-agent cooperation for planning, exchanging ideas, and solving problems.

Main Objectives

- ❏ Enable AI agents to communicate & reason collectively.
- ❏ Foster alignment with human goals & values.
- ❏ Provide a scalable framework for large agent societies.
- ❏ Serve as a platform for multi-agent research & applications.

What’s New in CAMEL?

Role-Playing Society

AI agents that think and act autonomously

Novel Inception Prompting

New prompting technique to keep conversations aligned and purposeful

Massive Scale

Generated 25,000+ role-playing dialogues to study emergent behaviors

Open Research Tools

Open tools to explore emergence, alignment, and cooperation in AI

🏆*CAMEL-AI beats GPT-3.5 in both human and GPT-4 evaluations*

Design Principles

Code-as-Prompt

Uses code and comments as prompts for agents.

Evolvability

Enables continuous system evolution through data and interaction.

Statefulness

Allows agents to maintain memory for complex tasks.

Scalability

Supports large-scale systems with efficient coordination.

Core Components

Agents

Autonomous LLM-powered Reasoners (e.g., ChatAgent, CriticAgent).

Agent Societies

Frameworks for multi-agent collaboration, planning, and critique via role-playing.

Interpreters & Tools

Execute code & integrate external functions for task automation.

Memory & Storage

Persistent context management and storage systems (e.g., key-value, vector, graph).

Retrievers & RAG

Knowledge retrieval pipelines for grounded and accurate responses.

Synthetic Data Engines

Pipelines for CoT, Self-Instruct, and dual-agent dataset generation.

Simulation Runtime

Environments for orchestrating large-scale multi-agent systems.

How They Work Together

Role Coordination

User → Societies ↔ Agents

Societies consists of agents (AI User, AI Assistant, Critic Agent, etc.).
Collaborate via role-driven conversations for task planning.

Reasoning & Execution

Agents ↔ RAG Pipelines ↔ Memory & Storage ↔ Interpreters/Tools

Agents retrieve knowledge, maintain context, and execute tasks.
Interpreters enable real-world action (API calls, Python scripts, etc.).

Data Generation Pipelines

Synthetic Data Engines (Self-Instruct, CoT, Self-Improve CoT)

Create training datasets and improve agent reasoning.

Simulation Runtime

Orchestrates multi-agent societies and workflows at scale (e.g., OASIS, CRAB).

Main Agent Architecture

🙋🏻‍♂️ **User**
Initial request/ query

↓

🤖 **Task Specifier Agent**
Refine ideas into tasks

↓

AI User
Give instructions & plan executions

AI Assistant
Follow instructions & execute tasks

Critic Agent
Select options & provide feedbacks

🏆 **Task Complete!**

🛠️ Role-Playing Framework

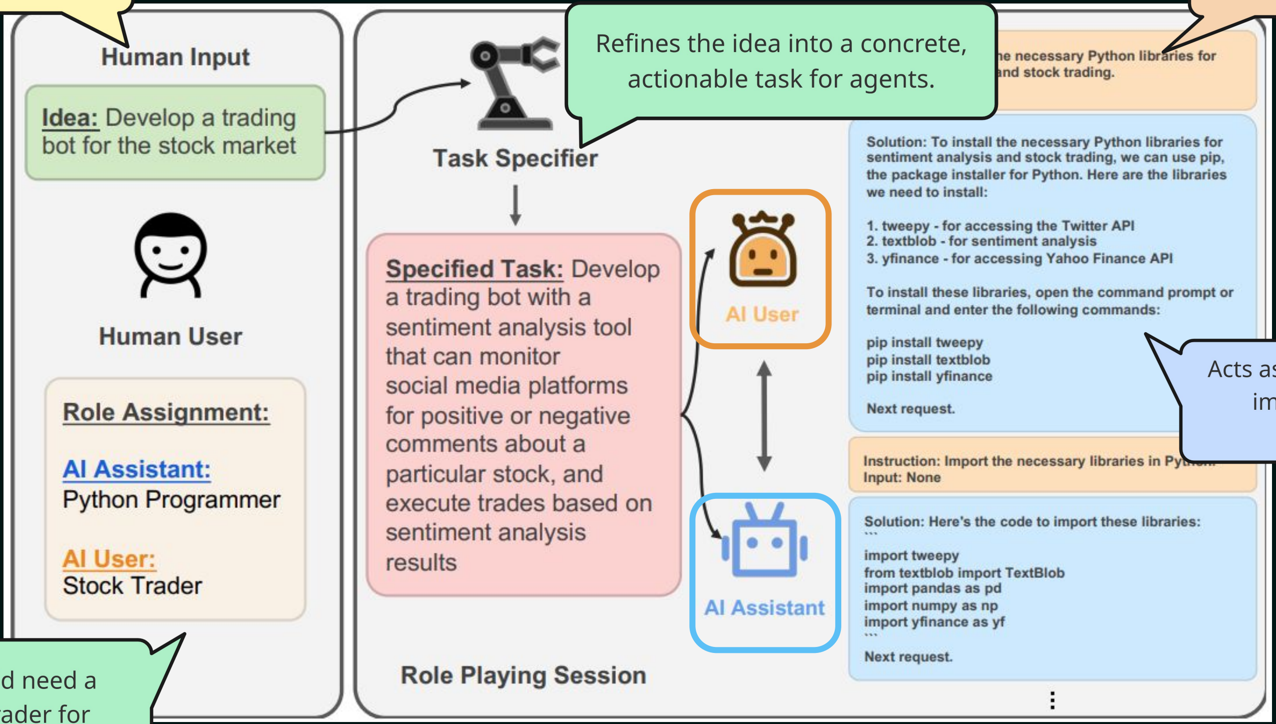
- ❑ **Main architecture** first introduced.
- ❑ Agents collaborate through **role-driven conversations:**
  - ❑ Human **User** provides an idea.
  - ❑ **Task Specifier Agent** refines it into a clear task.
  - ❑ Roles like **AI User** and **AI Assistant** are assigned.
  - ❑ **Critic Agent** provide feedbacks.
- ❑ Conversation follows **strict protocols** to avoid role flipping and infinite loops.
- ❑ Agents **interact autonomously** until task completion.

Role Playing in Action

A human provides a high-level idea "Develop a trading bot for the stock market."

Acts as the Stock Trader, planning steps and giving instructions to AI Assistant.

Refines the idea into a concrete, actionable task for agents.



Acts as the Python Programmer, implementing solutions as directed.

In real world, we would need a Programmer and a Trader for technical and domain expertise.

Figure 1: CAMEL Role-Playing Framework

✅ **Task completed without human intervention**

Inception Prompting

- Special system prompts applied at the start for task specification and role assignment.
- Task Specifier Prompt → *Refines vague ideas into clear tasks* for the Task Specifier Agent.
- Assistant System Prompt → *Assigns assistant role and sets rules "Never flip roles" and "Always start with Solution"*
- User System Prompt → *Defines user role and sets rules for instruction behaviors, like using <CAMEL_TASK_DONE> to signal task completion.*
- Prevents role-flipping, infinite loops, and flake replies.

AI Society Inception Prompt

Task Specifier Prompt:

Here is a task that <ASSISTANT_ROLE> will help <USER_ROLE> to complete: <TASK>.
Please make it more specific. Be creative and imaginative.
Please reply with the specified task in <WORD_LIMIT> words or less. Do not add anything else.

Assistant System Prompt:

Never forget you are a <ASSISTANT_ROLE> and I am a <USER_ROLE>. Never flip roles! Never instruct me!
We share a common interest in collaborating to successfully complete a task.
You must help me to complete the task.
Here is the task: <TASK>. Never forget our task!
I must instruct you based on your expertise and my needs to complete the task.

I must give you one instruction at a time.
You must write a specific solution that appropriately completes the requested instruction.
You must decline my instruction honestly if you cannot perform the instruction due to physical, moral, legal reasons or your capability and explain the reasons.
Unless I say the task is completed, you should always start with:

Solution: <YOUR_SOLUTION>

<YOUR_SOLUTION> should be specific, and provide preferable implementations and examples for task-solving.
Always end <YOUR_SOLUTION> with: Next request.

User System Prompt:

Never forget you are a <USER_ROLE> and I am a <ASSISTANT_ROLE>.
Never flip roles! You will always instruct me.
We share a common interest in collaborating to successfully complete a task.
I must help you to complete the task.
Here is the task: <TASK>. Never forget our task!
You must instruct me based on my expertise and your needs to complete the task ONLY in the following two ways:

1. Instruct with a necessary input:
Instruction: <YOUR_INSTRUCTION>
Input: <YOUR_INPUT>
2. Instruct without any input:
Instruction: <YOUR_INSTRUCTION>
Input: None

The "Instruction" describes a task or question. The paired "Input" provides further context or information for the requested "Instruction".

You must give me one instruction at a time.
I must write a response that appropriately completes the requested instruction.
I must decline your instruction honestly if I cannot perform the instruction due to physical, moral, legal reasons or my capability and explain the reasons.
You should instruct me not ask me questions.
Now you must start to instruct me using the two ways described above.
Do not add anything else other than your instruction and the optional corresponding input!
Keep giving me instructions and necessary inputs until you think the task is completed.
When the task is completed, you must only reply with a single word <CAMEL_TASK_DONE>.
Never say <CAMEL_TASK_DONE> unless my responses have solved your task.

Figure 2: Pre-defined System Prompts

Prompt Engineering in Action

Create a 2-day travel itinerary for New York based on real-time weather data for the upcoming weekend...

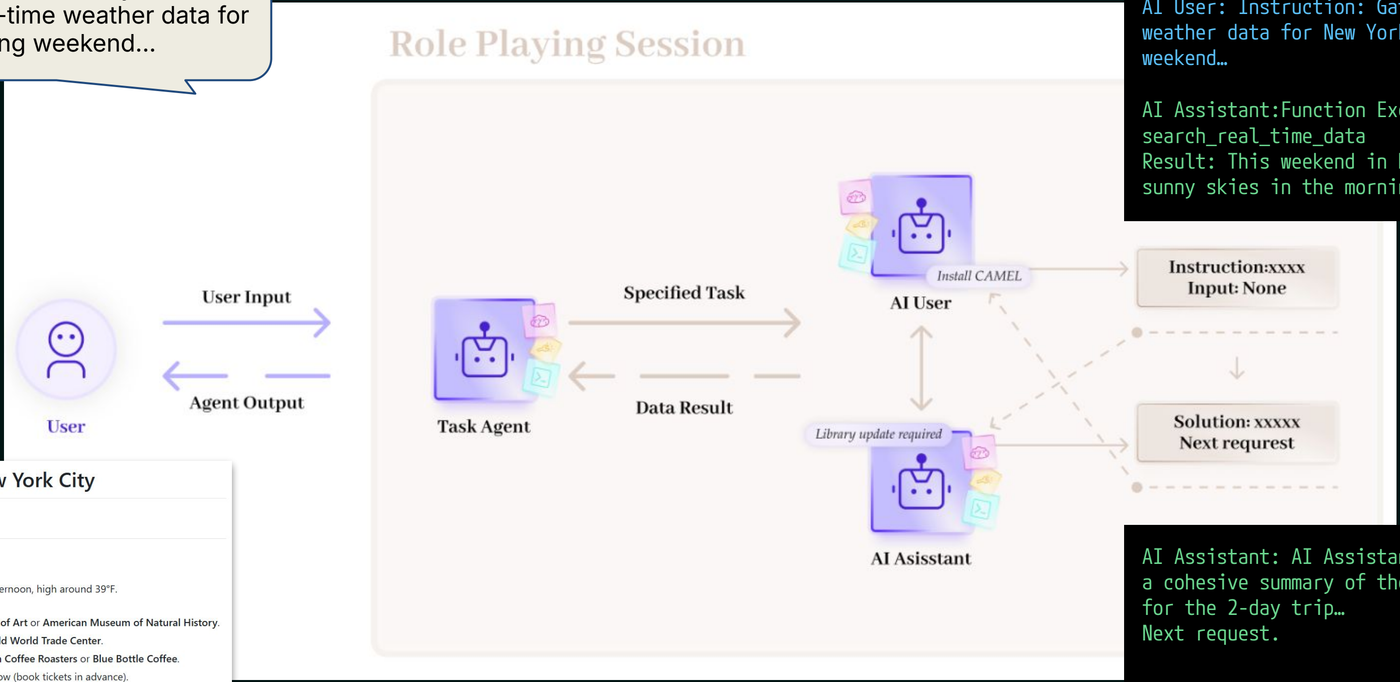


Figure 3: Role Playing System Design

```
AI User: Instruction: Gather real-time
weather data for New York for the upcoming
weekend…

AI Assistant:Function Execution:
search_real_time_data
Result: This weekend in New York City, expect
sunny skies in the morning on Saturday…
```

2-Day Travel Plan for New York City

Itinerary Overview

Saturday

- **Weather**: Sunny in the morning, overcast by afternoon, high around 39°F.
- **Key Activities**:
  - **Morning**: Visit **The Metropolitan Museum of Art** or **American Museum of Natural History**.
  - **Late Morning**: Indoor shopping at **Westfield World Trade Center**.
  - **Lunch**: Cozy café experience at **Stumptown Coffee Roasters** or **Blue Bottle Coffee**.
  - **Afternoon**: Attend a matinee Broadway show (book tickets in advance).
  - **Evening**: Dinner at **Chelsea Market** and optional brisk walk on the High Line (dress warmly).

Sunday

- **Weather**: Partly cloudy, high around 29°F, windy with WNW winds at 20 to 30 mph.
- **Key Activities**:
  - **Morning**: Enjoy a cozy brunch at **Balthazar** or **Sarabeth's** (make reservations).

```
AI Assistant: AI Assistant:Solution: Here is
a cohesive summary of the entire travel plan
for the 2-day trip…
Next request.


AI User:
<CAMEL_TASK_DONE>
```

Workforce Architecture

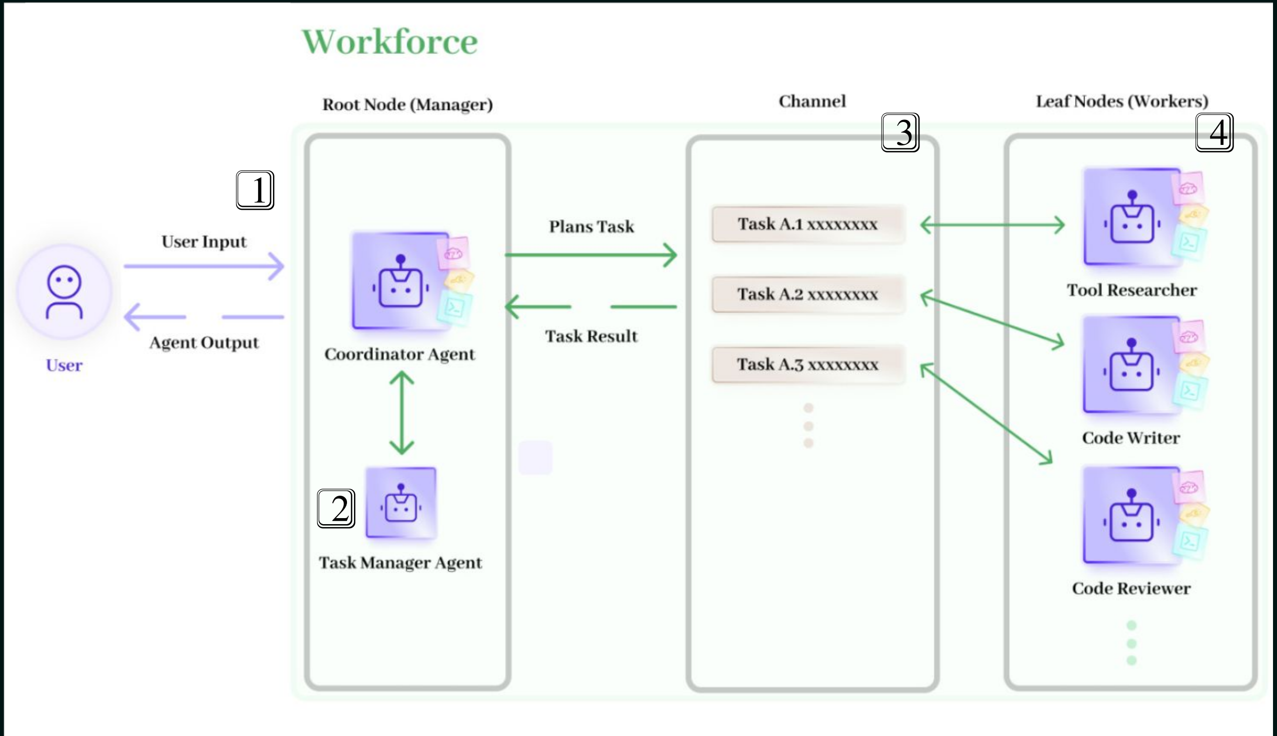


Figure 4: Workforce System Design

Workforce uses a hierarchical, modular design to solve tasks.

- ❑ Task Reception – Coordinator "project manager" receives high-level task
- ❑ Decomposition – Task Manager "strategy lead" splits big jobs into doable subtasks & controls workflow.
- ❑ Distribution – Coordinator assigns subtasks to workers (Researcher, Coder, Reviewer)
- ❑ Collaboration – Workers execute and share results
- ❑ Error Handling – Split or add agents
- ❑ Completion – Results are aggregated and returned

Workforce in Action

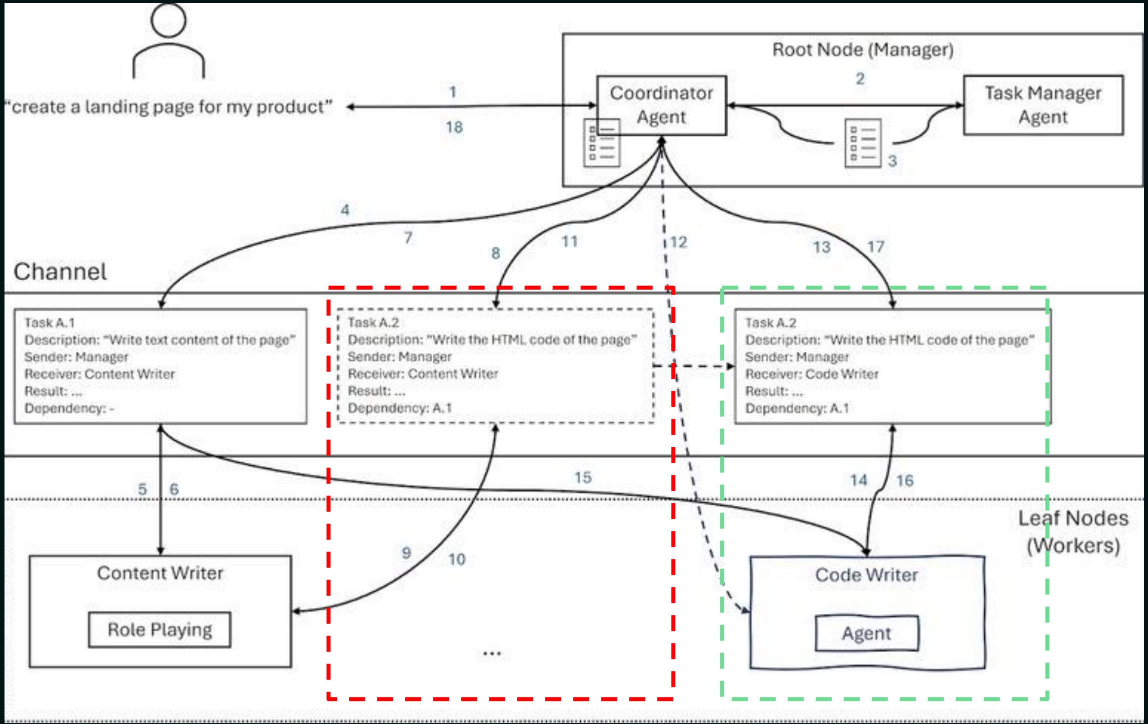


Figure 5: How Workforce Works

- ❏ Manager detects failure when Content Writer cannot handle a coding task.
- ❏ Dynamically creates a new Code Writer agent and assigns the task.
- ❏ Adapts on the fly to ensure all subtasks are completed fully autonomously.

Human-in-Loop Design

01

Set agent up

Define agent goals, tools & workflow parameters.

02

Agent does task

Agent runs autonomously until input is needed.

03

Get feedback

HumanLayer prompts user for approval or rejection.

04

Human checks

User approves or rejects actions via console or platform.

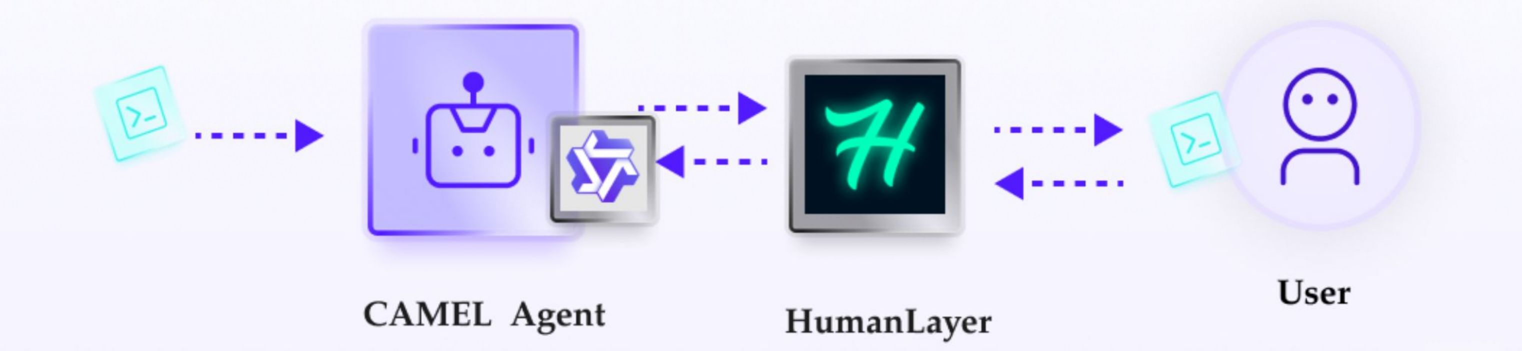
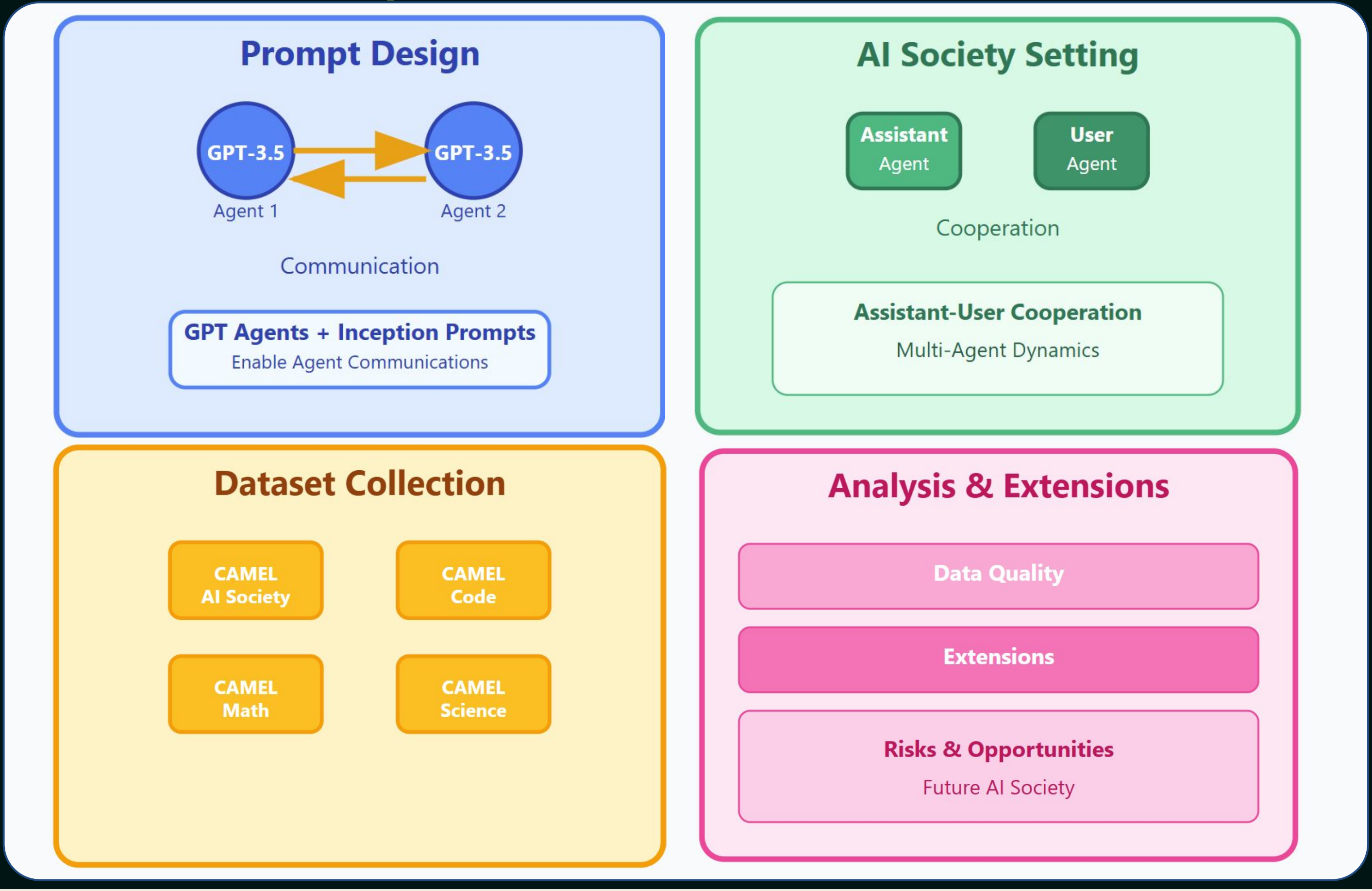


Figure 6: Human-in-Loop and Approval

Experiments Overview



Challenges & Solutions

Datasets Generated by CAMEL

- **AI Society:** 25,000 role-playing conversations (50×50 roles ×10 tasks)
- **Code:** Programming tasks across multiple domains
- **Math & Science:** 90,000 problem–solution pairs

These datasets are used in training and evaluating LLMs on reasoning and task solving.

CAMEL Evaluation & Results

Experimentation Setup

📂 *AI Society*: 100 reasoning tasks; **Human & GPT-4 evaluation**.

📂 *Code*: 100 programming tasks; **GPT-4 evaluation only**.

🧪 Compared **CAMEL agents** against **GPT-3.5.**

Dataset	Evaluation Type	Draw	*gpt-3.5-turbo Wins*	CAMEL Agents Win
AI Society	Human Evaluation	13.3%	10.4%	76.3%
	GPT4 Evaluation	4.0%	23.0%	73.0%
Code	GPT4 Evaluation	0.0%	24.0%	76.0%

Table 1: Agent Evaluation Results

✅ CAMEL wins 76.3% of human votes for AI Society tasks.

✅ GPT-4 evaluation shows CAMEL wins **73%** (AI Society) and **76%** (Code tasks).

✅ Shows **multi-agent cooperation** outperforms **single-shot generation**.

Further Experiments

Experiment Setup

- Start with **LLaMA-7B** (7B parameter base model).
- Fine-tune it progressively on CAMEL datasets.
- Compare two models:
  - **Model 1** = before adding a dataset
  - **Model 2** = after adding a dataset.
- Evaluate responses using **GPT-4**.

Dataset	Model 1 AI Society	Model 1 Code	Model 1 Math	Model 1 Science	Model 2 AI Society	Model 2 Code	Model 2 Math	Model 2 Science	Draw	Model 1	Model 2
AI Society					✓				0	6	**14**
Code					✓				0	0	**20**
Math					✓				**9**	5	6
Science					✓				0	13	**47**
AI Society	✓				✓	✓			4	8	**8**
Code	✓				✓	✓			1	9	**10**
Math	✓				✓	✓			5	**8**	7
Science	✓				✓	✓			1	19	**40**
AI Society	✓	✓			✓	✓	✓		5	6	**9**
Code	✓	✓			✓	✓	✓		1	9	**10**
Math	✓	✓			✓	✓	✓		1	3	**16**
Science	✓	✓			✓	✓	✓		3	8	**49**
AI Society	✓	✓	✓		✓	✓	✓	✓	3	1	**16**
Code	✓	✓	✓		✓	✓	✓	✓	1	8	**11**
Math	✓	✓	✓		✓	✓	✓	✓	**10**	5	5
Science	✓	✓	✓		✓	✓	✓	✓	9	2	**49**
AI Society					✓	✓	✓	✓	0	0	**20**
Code					✓	✓	✓	✓	0	0	**20**
Math					✓	✓	✓	✓	0	0	**20**
Science					✓	✓	✓	✓	0	0	**60**

Table 2: Emergence of Knowledge

Key Results

- Table 2 → Model 2 outperforms Model 1.
- Table 3 → CAMEL-7B solves 50% of coding tasks, and beats all other models, except gpt-3.5.

	HumanEval		HumanEval^{+}	
pass@k [%]	$k=1$	$k=100$	$k=1$	$k=100$
gpt-3.5-turbo	69.4	94.0	61.7	89.8
LLaMA-7B	10.5	36.5	-	-
Vicuna-7B	11.0	42.9	9.9	34.7
CAMEL-7B	**14.0**	**57.9**	**12.2**	**50.0**

Table 3: HumanEval(+) for Various Models

Strengths & Limitations

Strengths

- ❑ Enables autonomous multi-agent collaboration
- ❑ Produces high-quality instruction-following datasets
- ❑ Provides open-source tools for researchers
- ❑ Demonstrates emergent capabilities of LLMs
- ❑ Scales across domains (science, code, society)

Risks

- ❑ Agents may generate harmful or malicious plans (e.g., hacking scenario)
- ❑ Aligning AI with human values is still difficult

Limitations

- ❑ Evaluation depends on human preferences (possible bias)
- ❑ High cost for generating large datasets (API usage)

Future Work

- ❏ (At the time of publication) Scale from two agents to multi-agent societies
- ❏ Explore competitive as well as cooperative agent interactions
- ❏ Investigate embodied agents interacting with APIs and real-world tools
- ❏ Address challenges in AI alignment for large agent societies

💡 *As of 2025, the CAMEL team has already launched a multi-agent workforce module and started implementing these directions.*

Key Takeaways

CAMEL goes beyond research

It offers tools, datasets, and a flexible framework for multi-agent experimentation

An open-source ecosystem

With cookbooks, workflows, and GitHub support for community-driven innovation

Opportunities for contribution

Researchers can extend CAMEL, test use cases, and build new multi-agent paradigms

Shaping the future of AI collaboration

A blueprint for scaling agent societies while addressing alignment and safety challenges

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Thank You!